

Azure AI Certification Preparation Notes

1.1 Select the appropriate Azure AI services

Azure provides multiple AI services tailored for different business needs. - Azure OpenAI Service: Generative AI applications such as chatbots and content generation. - Azure AI Vision: Image recognition, OCR, and visual analysis. - Azure AI Language: NLP tasks including sentiment analysis and summarization. - Azure AI Speech: Speech recognition and synthesis. - Azure AI Document Intelligence: Information extraction from forms and invoices. - Azure AI Search: Knowledge mining and intelligent search solutions.

Key Exam Topics

- Generative AI solutions
- Computer vision solutions
- Natural language processing solutions
- Speech solutions
- Information extraction solutions
- Knowledge mining solutions

Practice Questions and Answers

Question	Answer
Q1. What is Azure OpenAI Service?	Azure OpenAI Service provides access to powerful language models such as GPT-4.
Q2. What is Computer Vision used for?	Computer Vision analyzes images and videos for object detection, OCR, and facial recognition.
Q3. What is NLP in Azure AI?	Natural Language Processing enables text analysis, translation, summarization, and sentiment analysis.
Q4. Why is Responsible AI important?	Responsible AI ensures fairness, transparency, accountability, privacy, and safety.
Q5. What is Azure AI Speech service?	Speech service provides speech-to-text, text-to-speech, translation, and speaker identification.
Q6. What is Azure AI Search?	Azure AI Search enables indexing and searching large datasets with AI enrichment capabilities.
Q7. Why use CI/CD with Azure AI?	CI/CD automates deployment, testing, and updates of AI applications.
Q8. How are Azure AI services secured?	Azure AI services use keys, managed identities, RBAC, and network security.
Q9. What is content moderation?	Content moderation detects and filters harmful or inappropriate content.
Q10. What are prompt shields?	Prompt shields protect AI systems from malicious prompts and jailbreak attempts.

1.2 Plan, create, and deploy an Azure AI service

Planning and deployment involve selecting suitable AI models, configuring Azure resources, and implementing DevOps practices. Key considerations: - Responsible AI principles - Resource provisioning in Azure Portal - Model deployment strategies - SDKs and APIs integration - CI/CD pipelines using GitHub Actions or Azure DevOps - Containerized deployments for edge or hybrid environments.

Key Exam Topics

- Responsible AI principles
- Create Azure AI resources
- Choose AI models
- Deploy AI models
- SDKs and APIs
- Default endpoints
- CI/CD integration
- Container deployment

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1.3 Manage, monitor, and secure an Azure AI service

Azure AI services require continuous monitoring and security management. Important practices: - Use Azure Monitor and Application Insights - Track usage and optimize costs - Rotate and secure API keys - Use Microsoft Entra ID authentication - Implement RBAC and network security rules.

Key Exam Topics

- Monitor Azure AI resources
- Manage costs
- Manage account keys
- Manage authentication

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1.4 Implement AI solutions responsibly

Responsible AI ensures safe and ethical AI systems. Best practices: - Configure content safety filters - Detect harmful or biased outputs - Apply moderation policies - Use prompt shields against prompt injection attacks - Establish AI governance and compliance frameworks.

Key Exam Topics

- Content moderation
- Responsible AI insights
- Content filters and blocklists
- Prompt shields and harm detection
- AI governance framework

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